IMD3002 – 3D COMPUTER GRAPHICS

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L18 – Textures II: Advanced Methods

Overview

- In this lecture we continue with texturing and look at some of the more advanced techniques available such as Displacement Maps, Parallax Maps, Light Maps, and Reflection Maps

Parallax Mapping

- Introduction
  - Parallax Mapping is an enhancement of bump mapping or normal mapping techniques
  - Sometimes called **Offset Mapping** or **Virtual Displacement Mapping**)
  - Parallax Mapping has the advantage over the other two because it provides much more of an apparent depth and realism to texture grain without significantly impacting rendering speed

Parallax Mapping

- Introduction
  - **Example:** Parallax Mapping in the game “F.E.A.R”; left (without) right (with)

Parallax Mapping

- Introduction
  - **Example:** Parallax Mapping in the game "Splinter Cell"; left (without), right (with)

Parallax Mapping

- Implementation
  - Parallax Mapping is a relatively new method; introduced in 2001
  - It is implemented by displacing the texture coordinates at a point on the rendered polygon by a function of the view angle in tangent space (the angle relative to the surface normal) and the value of the height map at that point
  - At steeper view angles the texture coordinates are displaced more, thus giving the illusion of depth due to parallax effects

Parallax Mapping

- Silhouettes and Shadows
  - The method is normally a single step process that does not account for occlusion
  - Subsequent enhancements have been made to the method which incorporate iterative approaches to allow for occlusion and accurate silhouette rendering
  - Parallax Mapping is not yet common in games as it is comparatively slow due to the need for so many samples and conditionals for each pixel

Relief Mapping

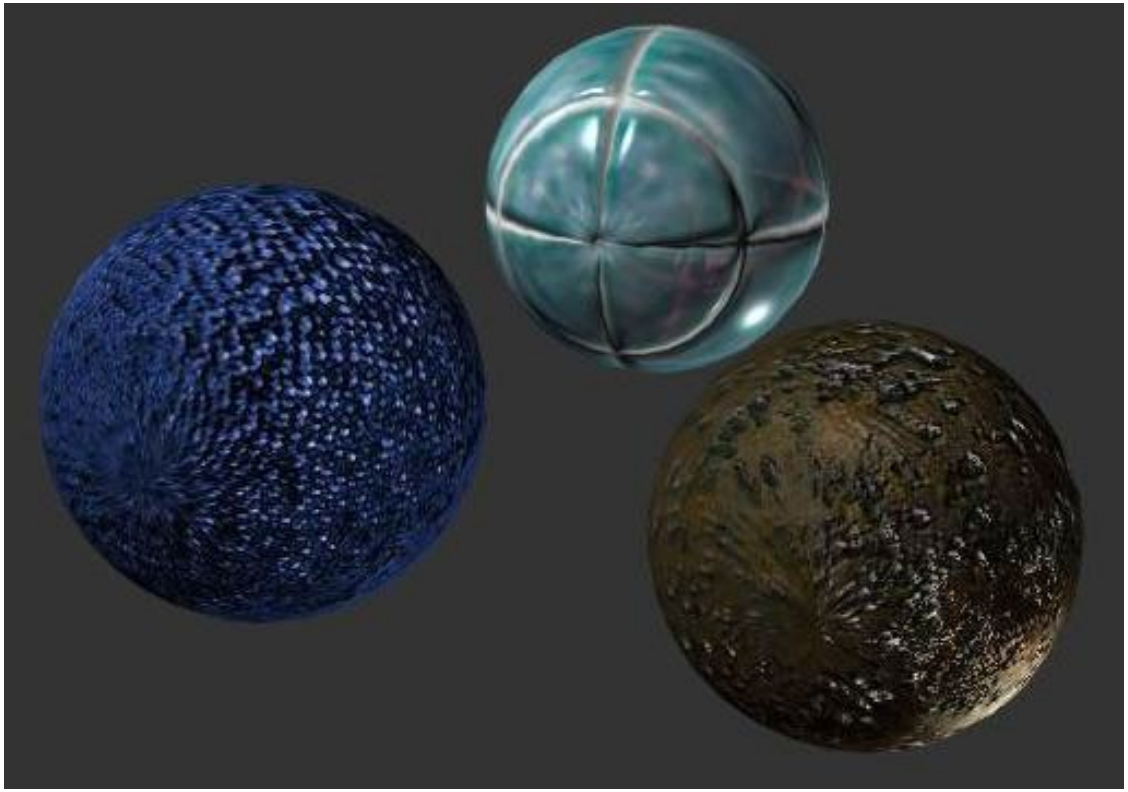
- Overview
  - Relief Mapping is an alternative to parallax mapping that is much more accurate, and can support self-shadowing and normal mapping
  - It involves offsetting texture coordinates towards the viewer, sampling a number of pixels in between, then drawing whichever one satisfies a certain height criteria

Displacement Mapping

- Introduction
  - Displacement Mapping is an alternative technique to bump mapping and its derivations (normal mapping and parallax mapping)
  - It uses a procedural texture or a height map to create the effect where the actual geometric position of points over the textured surface are displaced according to the value the texture function evaluates to each point on the surface

Displacement Mapping

- **Example:** bump mapped objects, although appearing to have a rough surface – the silhouette is still smooth

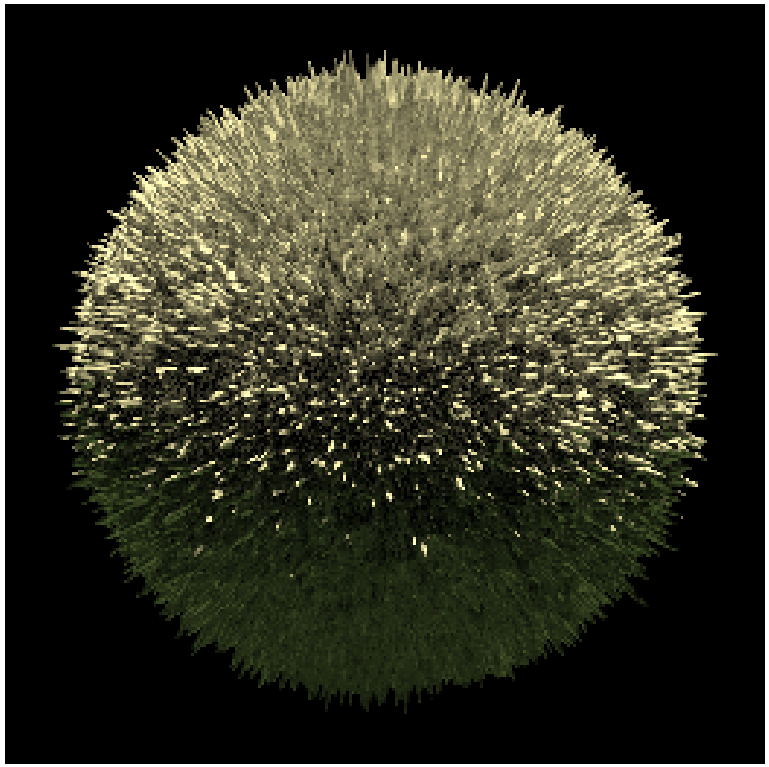


Displacement Mapping

- Introduction
  - It provides surfaces a great sense of depth and detail, permitting in particular self-occlusion, self-shadowing and silhouettes; on the other hand, it is the most costly of this class of techniques owing to the large amount of additional geometry
  - The key difference between ***displacement mapping*** and ***bump mapping*** is that with bump mapping the normal vector alone is perturbed – in displacement mapping the actual geometry is perturbed

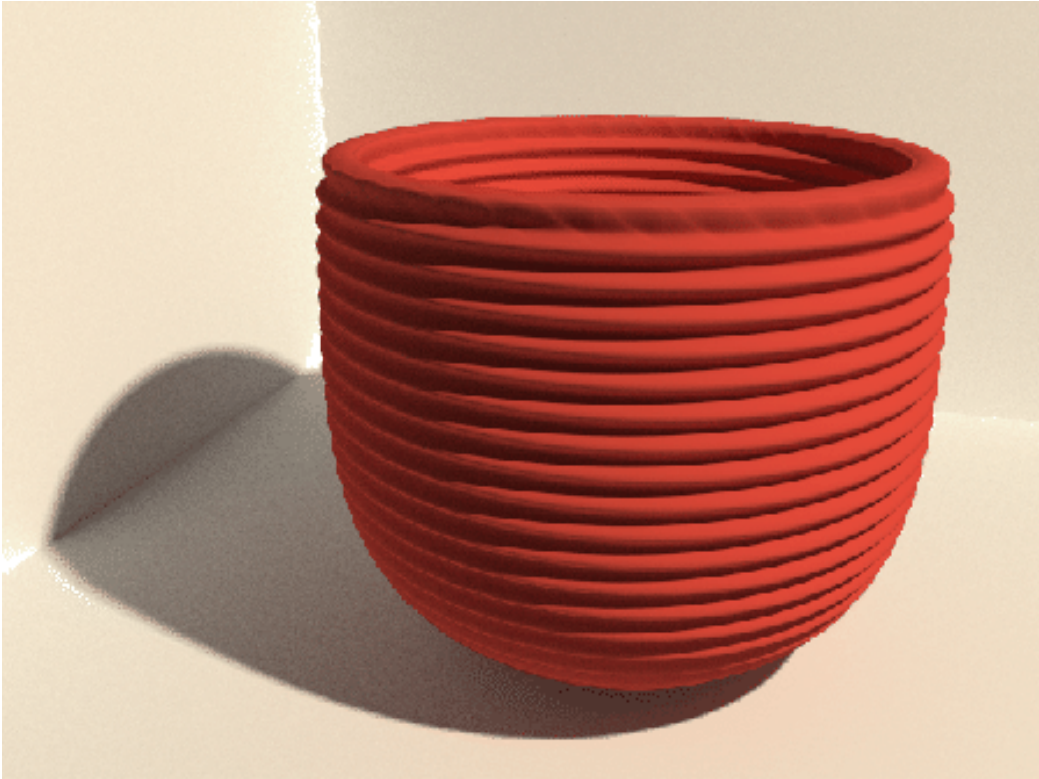
Displacement Mapping

- Introduction
  - **Example:** this is an icosahedron; the based polygonal mesh has only 20 triangles



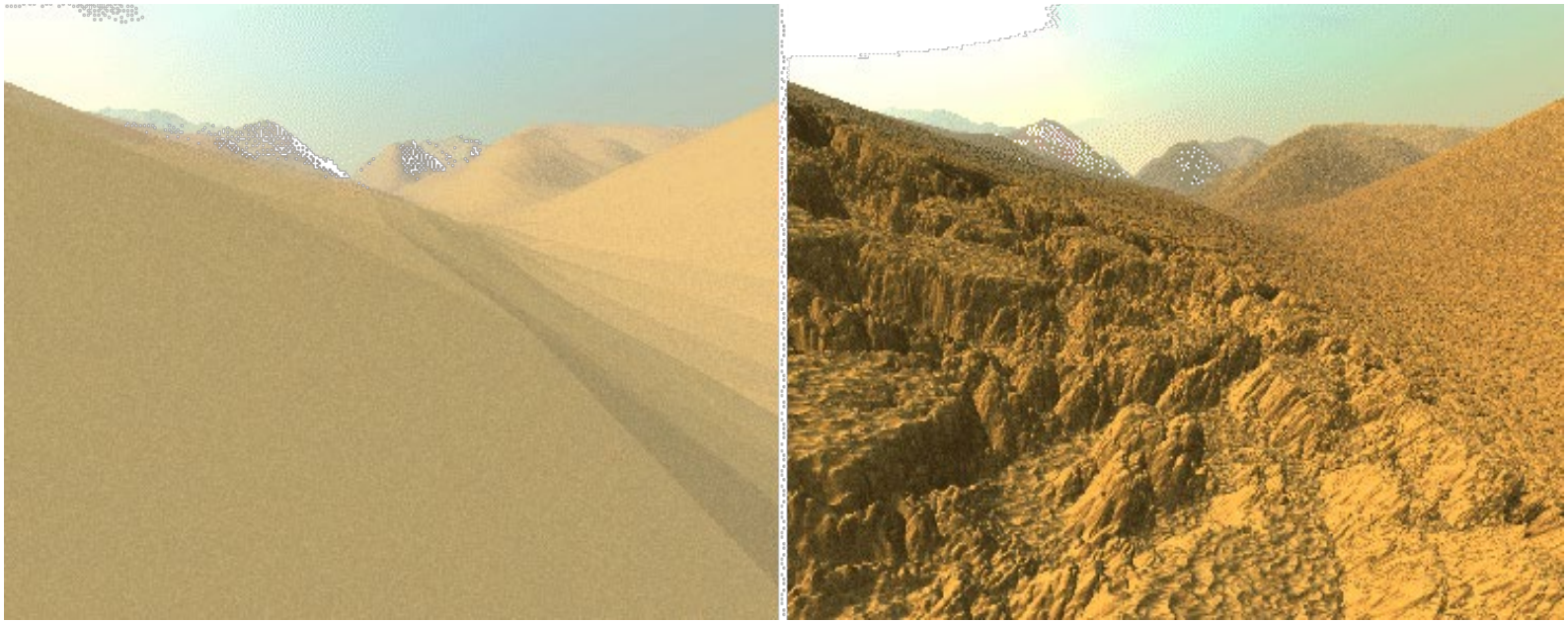
Displacement Mapping

- Introduction
  - **Example:** pottery with 4860 polygons; the geometry would normally require around 30M polygons



Displacement Mapping

- Introduction
  - **Example:** desert, without and with displacement
    - **Left image** ~ no displacement maps → 110K polygons
    - **Right image** ~ displacement maps (procedural), simulates ~1×10^{12} polygons



Displacement Mapping

- Introduction
  - The main advantage in using displacement mapping is that it is a routine that can be performed on the graphics card (i.e. the GPU) based on a texture
  - Therefore, we can pass a simple geometry (i.e. relatively few polygons) and a texture height map and the GPU can displace the geometry accordingly
  - The advantage to this is that passing same geometry down a computer's bus would be very slow as the geometry would be very large

Displacement Mapping

- Introduction
  - The other advantages to Displacement Mapping come in the obvious form of realism; with detailed geometry there is much greater realism
  - Compare the Bump Mapping techniques we have seen previously; we obtain problems with the silhouette of an object
  - At the edge of an object, even though it is supposed to be rough, because we didn't perturb the surface (only the normal) it is still smooth (when it should be rough)

Displacement Mapping

- Introduction
  - It has only been recently that displacement mapping can be used by DirectX and OpenGL – it was previously only used by systems such as RenderMan
  - One of the main reasons for this is that originally displacement mapping implementations required an adaptive tessellation of the surface in order to obtain the micro-polygons necessary (i.e. those which match the pixel resolution of the screen)

Displacement Mapping

- Introduction
  - The key to displacement maps is the use of either a height map (which is usually defined explicitly) or a procedural texture, which is defined using an implicit method
  - Procedural textures are often called 3D textures or 3D texture mapping and are described in the following slides

Displacement Mapping

- 3D Texture Mapping
  - 3D texture maps are defined in three dimensions, T(x,y,z), and are mapped onto a three dimensional object
  - For each point in three dimensional space we must find the corresponding point in the 3D texture map space and colour the 3D point with the value from the texture map

Displacement Mapping

- 3D Texture Mapping
  - 3D texture maps are generally not stored, because they would be too large, and so are procedurally defined and computed when needed
  - Thus, this method is frequently called procedural texturing
  - Note that for ray traced images, the intersection point is the 3D world point to use in the procedural texture

Displacement Mapping

- 3D Texture Mapping
  - A procedural texture is a generated image created using an algorithm intended to produce a realistic representation of natural elements such as wood, marble, granite, metal, stone, and others.
  - Usually, the natural look of the rendered result is achieved by the usage of fractal noise and turbulence functions
  - These functions are used as a numerical representation of the “randomness” found in many natural items

Displacement Mapping

- 3D Texture Mapping
  - In general, these noise and fractal functions are simply used to "*disturb*" the texture in a natural way such as the undulations of the veins of the wood
  - In other cases, like marble textures, they are based on the graphical representation of fractal noise

Light Maps

- Introduction
  - A light map is a texture map applied to a material to simulate the effect of a local light source
  - Like specular highlights, it can be used to improve the appearance of local light sources without resorting to excessive tessellation of the objects in the scene

Light Maps

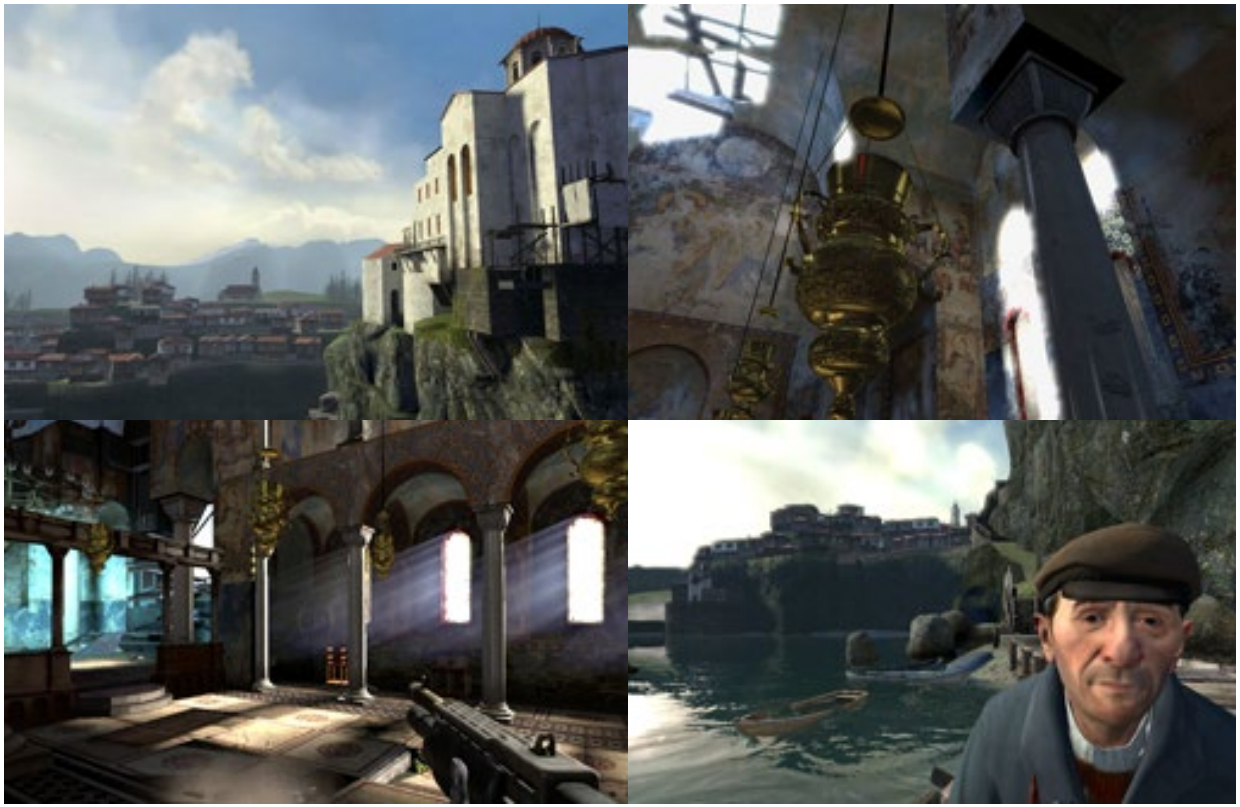
- Introduction
  - **Example:** simple scene that can be rendered in real-time, but with nice lighting due to light maps

Light Maps

- Introduction
  - **Example:** A Quake3-level rendered using multiple texture stages for light maps

Light Maps

- Introduction
  - **Example:** HDR Light Maps from Half-Life 2 “The Lost Coast”



Light Maps

- Introduction
  - One of the best examples using light maps, which we have just seen previously, is Half-Life 2
  - This game uses light maps to simulate the effects of local light sources, both stationary and moving, to very realistic effect

Light Maps

- Multi-Pass Sampling
  - Using light maps usually requires a multi-pass algorithm, unless the objects being mapped are not textured
  - A texture simulating the light's effect on the object is created, then applied to one or more objects in the scene

Light Maps

- Multi-Pass Sampling
  - Appropriate texture coordinates are generated, and texture transformations can be used to position the light, and create moving or changing light effects
  - Multiple light sources can be generated with a combination of more complex texture maps and/or more passes to the algorithm

Light Maps

- 2D Texture Light Maps
  - A 2D light map is a texture map applied to the surfaces of a scene, modulating the intensity of the surfaces to simulate the effects of a local light
  - If the surface is already textured, then applying the light map becomes a multi-pass operation, modulating the intensity of a surface detail texture

Light Maps

- 2D Texture Light Maps
  - A 2D light map can be generated analytically, creating a bright spot in luminance or colour values that drop off appropriately with increasing distance from the light center
  - As with other lighting equations, a quadratic drop off, modified with linear and constant terms can be used to simulate a variety of lights, depending on the area of the emitting source

Light Maps

- 3D Texture Light Maps
  - 3D Textures can also be used as light maps. One or more light sources are represented in 3D data, then the 3D texture is applied to the entire scene
  - The main advantage of using 3D textures for light maps is that it's easy to calculate the proper texture coordinates. The textured light source can be positioned globally with the appropriate texture transformations then the scene is rendered

Light Maps

- Texture Baking
  - We can obtain light maps in various ways, either for multi-pass rendering (where the light map is separated) or single-pass rendering where there is no texture or the texture and the light map are combined
  - Maya, 3DSMax, Blender all have methods for “baking” textures
  - Here is an example of baking textures in Maya:
    - http://fromthehill.nl/tutorials/bake/index.html

Reflection Mapping

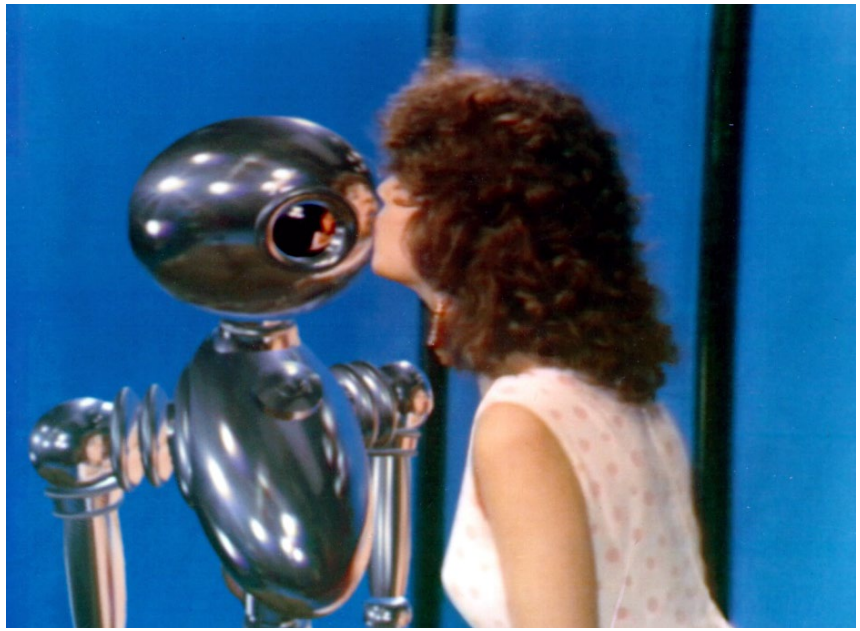
- Introduction
  - Reflection mapping is a method of simulating <u>some</u> of the effects of global illumination model using only scan line rendering
  - It provides us with an efficient method for simulating a mirrored surface by means of a pre-calculated texture image
  - The texture is used to store the image of the environment surrounding the rendered object

Reflection Mapping

- Introduction
  - **Example:** reflection mapping used in Pixar's "ShutterBug" test

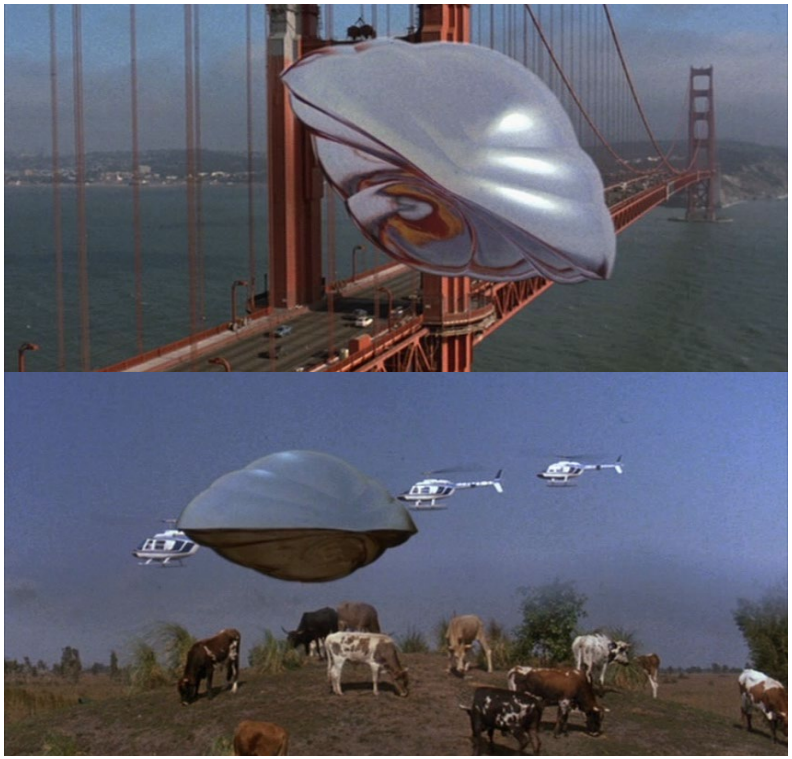
Reflection Mapping

- Introduction
  - **Example:** here is a shot from the film "***Interface***" (1984) where a woman is kissing a robot, in reality she is kissing a 10" sphere which is then reflection mapped onto the CG figure



Reflection Mapping

- Introduction
  - **Example:** here are some very famous shots from the film "***Flight of the Navigator***" (1986)



Reflection Mapping

- Introduction
  - **Example:** liquid metal terminator from "***Terminator 2: Judgement Day***" (1991)

Reflection Mapping

- Method Overview
  - First the scene is rendered from the view point of the surface to be mapped
  - Next, this image is turned into a texture map, which is then mapped to the surface
  - Note that this only does a "one bounce" reflection, whereas true global illumination algorithms, such as ray tracing, will do multiple bounces
  - A drawback of this is the absence of self reflections: we cannot see any part of the reflected object inside the reflection itself

Reflection Mapping

- Method Overview
  - There are several ways of storing the surrounding environment; the most common ones are:
    - **Spherical Environment Mapping** – where a single texture contains the image of the surrounding as reflected on a mirror ball; this is commonly known as "Standard Environment Mapping"
    - **Cubic Environment Mapping** – where the environment is unfolded onto the six faces of a cube and stored therefore as six square textures

Reflection Mapping

- Spherical Environment Mapping
  - This involves the use of a textured sphere at an infinite distance away from the object that reflects it
  - By creating a spherical texture using a fisheye lens, via pre-rendering, or using a light probe, the texture is mapped onto a hollow sphere, and the colours are calculated by using the light vectors from the points on the object to points on the environment map

Reflection Mapping

- Spherical Environment Mapping
  - This technique may produce results that are superficially similar to those produced by ray tracing, but incurs less of a performance hit because all of the colours of the points to be referenced are calculated beforehand
  - Therefore, all the renderer has to do is calculate the angles of incidence and reflection

Reflection Mapping

- Cubic Environment Mapping
  - There are a few key problems with Spherical Environment Mapping:
    - Due to the nature of the texture used for the map, there is an abrupt point of singularity on the rear of objects mapped onto a sphere
    - Cubic Environment Mapping addresses at least this issue

Reflection Mapping

- Cube Mapping
  - Since cube maps (if made and filtered correctly) have no visible seams, they are an obvious successor to Spherical Maps
  - Today, Spherical Environment Maps are almost nonexistent in certain modern graphical applications, such as games

Reflection Mapping

- Cube Mapping
  - Cube mapped reflection is a technique that uses cube mapping to make objects look like they reflect the environment around them
  - Generally, this is done with the same skybox that is used in outdoor renderings, as we have discussed previously
  - Though this is not a true reflection since objects around the reflective one will not be seen in the reflection, the desired effect is usually achieved

Reflection Mapping

- Cube Mapping
  - Cube mapped reflection is done by determining the vector that the object is being viewed at
  - This camera ray is reflected about the surface normal of where the camera vector intersects the object
  - This results in the reflected ray which is then passed to the cube map to obtain the effective pixel value that the camera then sees as if it is on the surface of the object
  - This creates the effect of a reflective object